**Example 4.21**

Find $\sin 105^\circ + \cos 105^\circ$

***Solution***

$\sin 105^\circ + \cos 105^\circ$

$= \sin(90^\circ + 15^\circ) + \cos 105^\circ$

$= \cos 15^\circ + \cos 105^\circ$

$= \cos 105^\circ + \cos 15^\circ$

$= 2\cos\left(\frac{105^\circ + 15^\circ}{2}\right)\cos\left(\frac{105^\circ - 15^\circ}{2}\right)$

$= 2\cos\left(\frac{120^\circ}{2}\right)\cos\left(\frac{90^\circ}{2}\right)$

$= 2\cos 60^\circ \cos 45^\circ = \frac{1}{\sqrt{2}}$

**Example 4.22**

Prove that

$$(\cos\alpha + \cos\beta)^2 + (\sin\alpha + \sin\beta)^2 = 4\cos^2\left(\frac{\alpha - \beta}{2}\right)$$

***Solution***

$\cos\alpha + \cos\beta =$

$$2\cos\left(\frac{\alpha + \beta}{2}\right)\cdot\cos\left(\frac{\alpha - \beta}{2}\right) \quad \ldots.(1)$$

$\sin\alpha + \sin\beta =$

$$2\sin\left(\frac{\alpha + \beta}{2}\right)\cdot\cos\left(\frac{\alpha - \beta}{2}\right) \quad \ldots.(2)$$

Squaring and adding (1) and (2)

$(\cos\alpha + \cos\beta)^2 + (\sin\alpha + \sin\beta)^2$

$= 4\cos^2\left(\frac{\alpha + \beta}{2}\right)\cos^2\left(\frac{\alpha - \beta}{2}\right) +$

$4\sin^2\left(\frac{\alpha + \beta}{2}\right)\cos^2\left(\frac{\alpha - \beta}{2}\right)$

$= 4\cos^2\left(\frac{\alpha - \beta}{2}\right)\left[\cos^2\left(\frac{\alpha + \beta}{2}\right)\right.$

$\left. + \sin^2\left(\frac{\alpha + \beta}{2}\right)\right]$

$= 4\cos^2\left(\frac{\alpha - \beta}{2}\right)$

## Exercise 4.3

1. Express each of the following as the sum or difference of sine or cosine:

   (i) $\sin\frac{A}{8}\sin\frac{3A}{8}$

   (ii) $\cos(60^\circ + A)\sin(120^\circ + A)$

   (iii) $\cos\frac{7A}{3}\sin\frac{5A}{3}$

   (iv) $\cos 7\theta \sin 3\theta$

2. Express each of the following as the product of sine and cosine

   (i) $\sin A + \sin 2A$

   (ii) $\cos 2A + \cos 4A$

   (iii) $\sin 6\theta - \sin 2\theta$

   (iv) $\cos 2\theta - \cos\theta$

3. Prove that

   (i) $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$

   (ii) $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$

4. Prove that

   (i) $(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2$

   $= 4\sin^2\left(\frac{\alpha - \beta}{2}\right)$

   (ii) $\sin A \sin(60^\circ + A)\sin(60^\circ - A)$

   $= \frac{1}{4}\sin 3A$

5. Prove that

   (i) $\sin(A - B)\sin C + \sin(B - C)$

   $\sin A + \sin(C - A)\sin B = 0$

   (ii) $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13}$

   $+ \cos\frac{5\pi}{13} = 0$



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